

## Operating System Design & Implementation

### Design Goals:

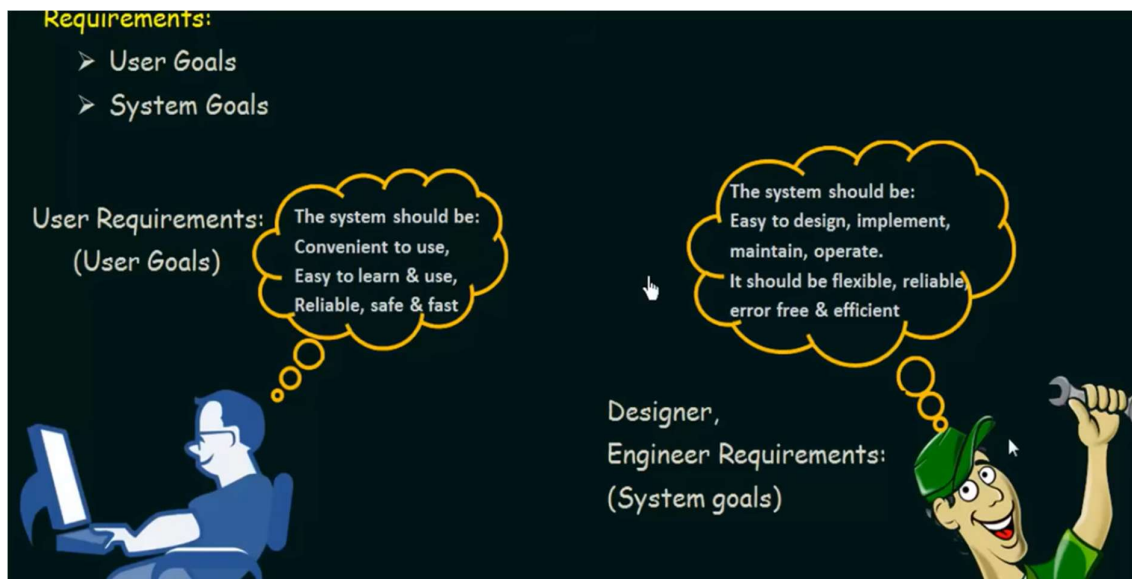
#### 1<sup>st</sup> Problem: - Defining Goals and specification

- Choice of Hardware
- Type of System

Beyond this highest design level, the requirements may be much harder to specify.

### Requirements:

- User Goals
- System Goals

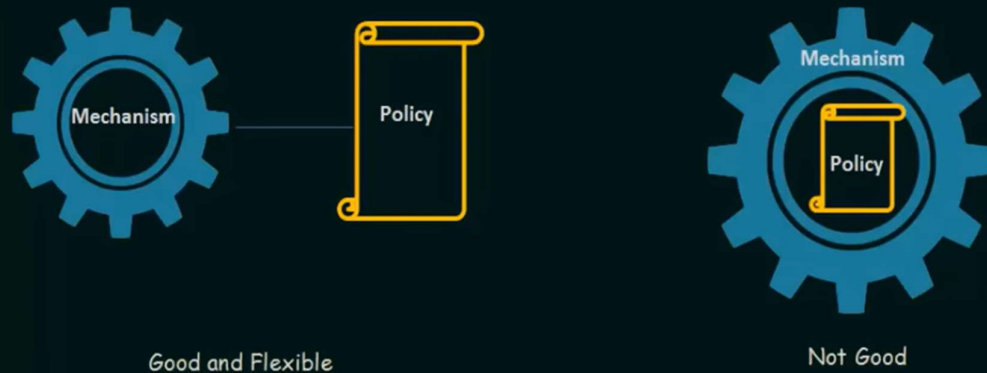


### Mechanisms and Policies:

Mechanisms determine how to do something.

Policies determine what will be done.

**One important principle is the separation of policy from mechanism.**



### Implementation:

- Once an operating system is designed, it must be implemented.
- Traditionally, operating systems have been written in assembly language.
- Now, however, they are most commonly written in higher-level languages such as C or C++

Advantages of writing in high level languages:

- The code can be written faster
- It is more compact
- It is easier to understand and debug
- It is easier to port

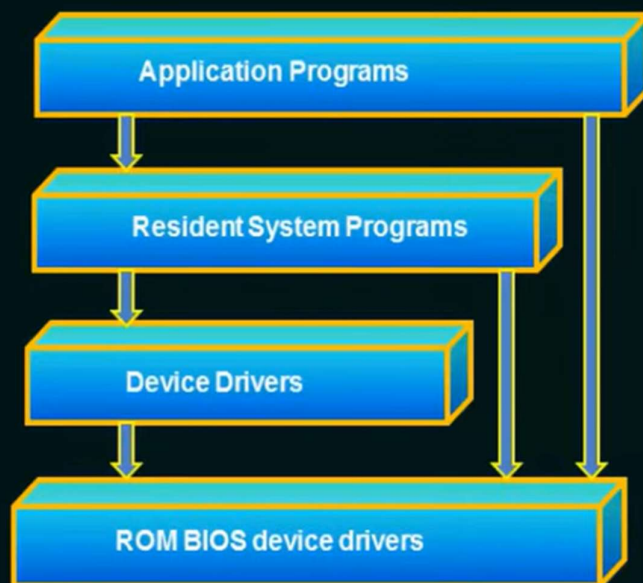
E.g.

MS-DOS was written in Intel 8088 assembly language. Consequently, it is available on only the Intel family of CPUs.

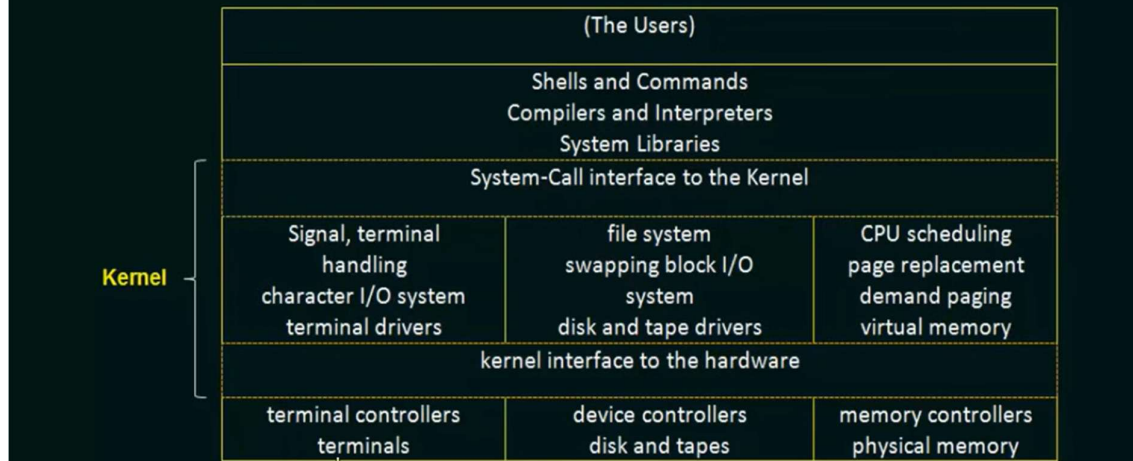
The Linux operating system, in contrast, is written mostly in C and is available on a number of different CPUs, including Intel 80X86, Motorola 680X0, SPARC, and MIPS RX000.

## Structures of Operating System

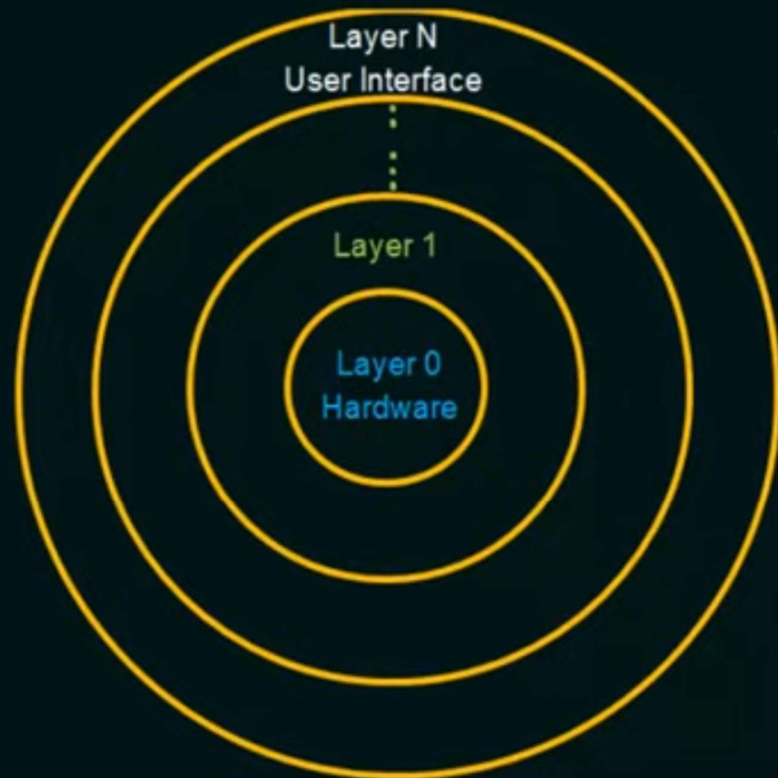
### Simple Structure



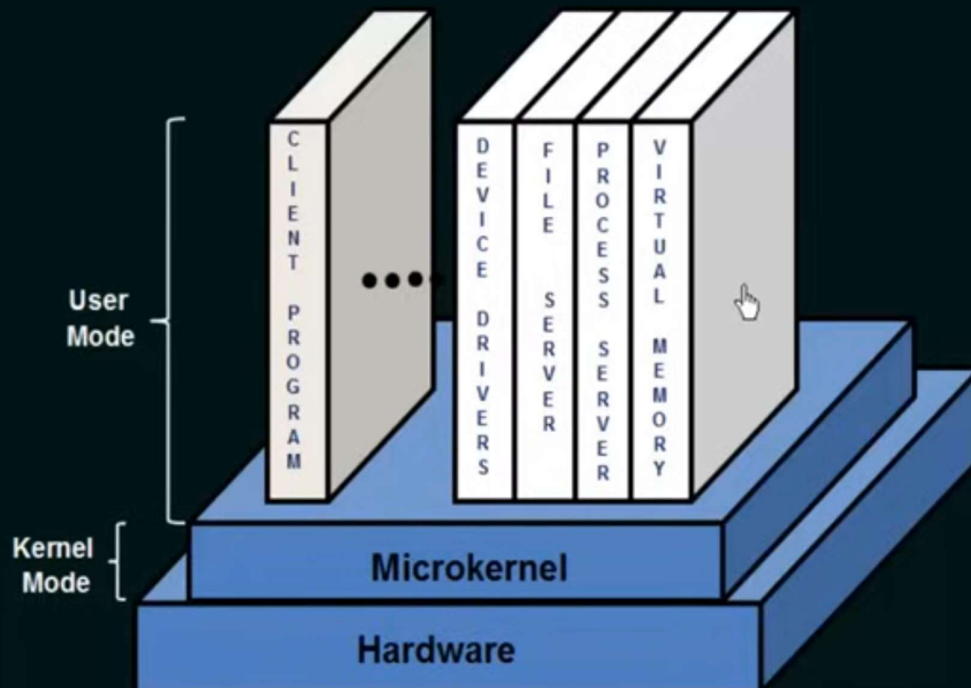
## Monolithic Structure



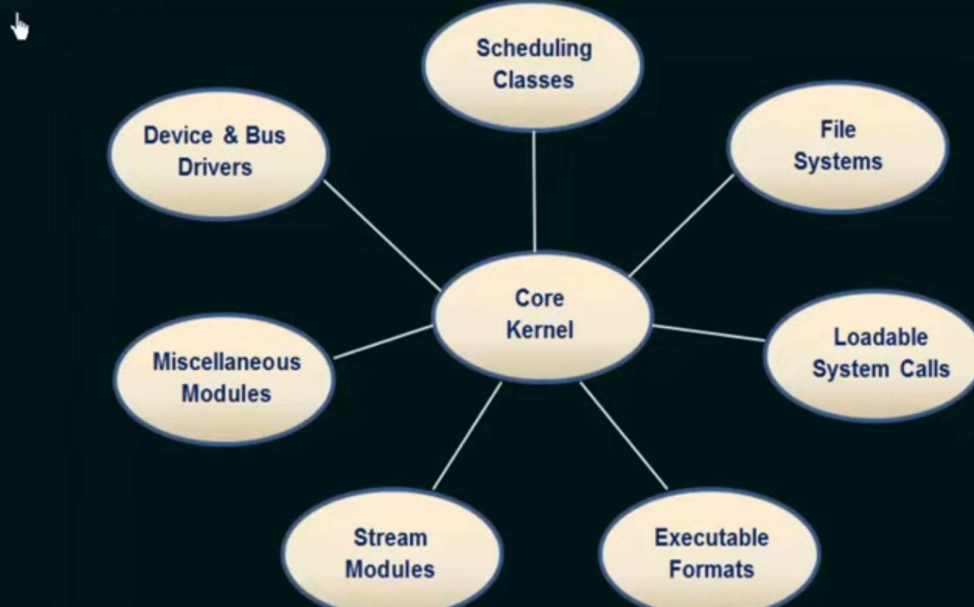
## Layered Structure



## Microkernels

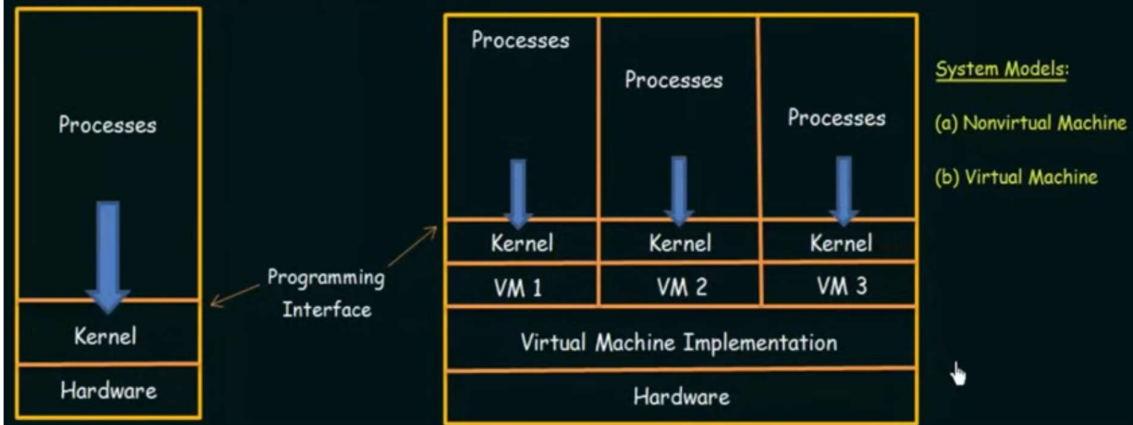


## Modules



## Virtual Machines

The fundamental idea behind a virtual machine is to abstract the hardware of a single computer (the CPU, memory, disk drives, network interface cards, and so forth) into several different execution environments, thereby creating the illusion that each separate execution environment is running its own private computer.



> It is a software based computer that runs inside a real computer.

### → Why Virtual Machine?

- Multiple OS on one System
- Safe Testing Environment
- Learning New Operating Systems
- Better Hardware Utilization
- Isolation and Security

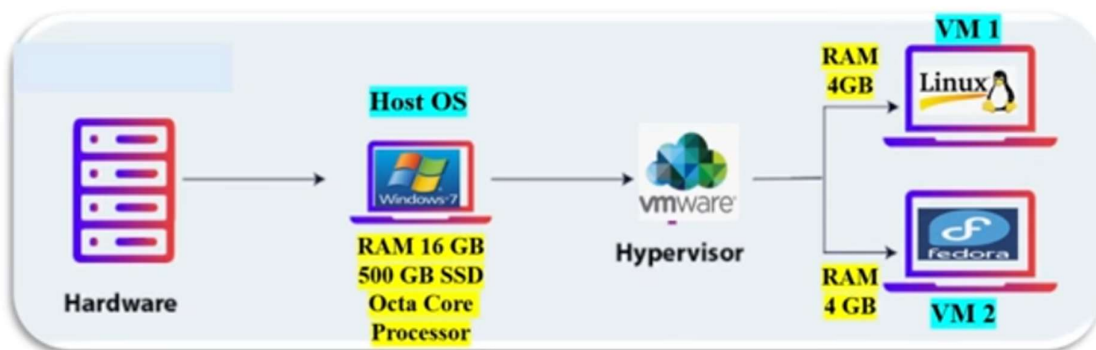
## Components of Virtual Machine

1. Host Machine : The real, physical computer (*Your laptop - Windows OS*).
2. Guest Machine : The virtual machine running inside host (*Ubuntu VM*).
3. Hypervisor – The software that manages VMs. (*VirtualBox or Vmware*)

## Hypervisor:

- It is a special software that allows you to create and run multiple virtual machines (VMs) on a single physical computer.
- It acts like a manager or controller between physical OS & virtual OS.

➤ **Popular Hypervisors:** VMware, KVM, Oracle VM VirtualBox, Microsoft Hyper-V etc.



### IMPLEMENTATION

Virtual Machine Software -  
Virtual Machine itself -

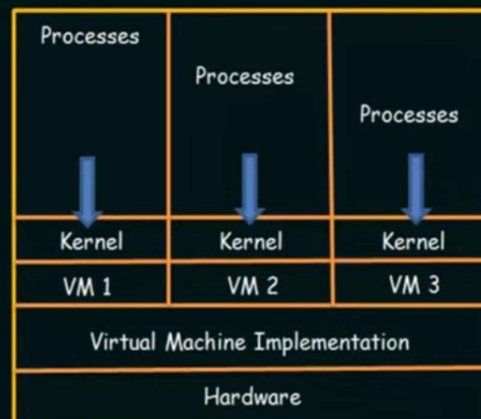
Runs in Kernel mode  
Runs in User Mode

Just as the physical machine has two modes, however,  
so must the virtual machine.

Consequently, we must have:

- A virtual user mode and
- A virtual kernel mode

BOTH OF WHICH RUN IN A PHYSICAL USER MODE



## Types of Virtual Machine

| <b>1. System Virtual Machine</b>                                                         | <b>2. Process Virtual Machine</b>                                                     | <b>3. Cloud based Virtual Machine</b>                                                                            |
|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>A System VM emulates a complete physical computer.</b><br><i>VirtualBox or Vmware</i> | <b>A Process VM is designed to run one application or program.</b><br><i>JVM, PVM</i> | <b>Run on remote infrastructure provided by cloud service providers.</b><br><i>Azure VM, AWS EC2, Google GCE</i> |

## Applications of Virtual Machines

- 1. Software Testing & Development**
- 2. Cybersecurity & Ethical Hacking**
- 3. Learning and Training**
- 4. Running Legacy Software**
- 5. Web & App Hosting (Cloud Computing)**
- 6. Disaster Recovery & Backup**
- 7. Safe Browsing & Internet Isolation**